AI TrustLens

An AI Output Auditing Platform Using NER and Graph Database

BTech Major Project First Review

Team No. 19 Members

- Aadhithyan P.S. – LNSS23CS072
- Fahad M.K. – LNSS23CS074
- Prajith S – LNSS23CS076
- Sneha Nair – NSS23CS060

Guided By

Assistant Professor: Abhishek S Kumar

Department of Computer Science and Engineering

NSS College of Engineering

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1. Introduction

- **AI-generated responses** can be useful but may contain incorrect, fabricated, or unsupported information.
- **Hallucinations** reduce the reliability and trustworthiness of AI outputs.
- **Verification challenges:** unstructured evidence and limited explicit representation of entity relationships.
- **Need:** claim-level, evidence-backed auditing with transparent and traceable results.
- **AI TrustLens** combines NER, RAG, and graph-based context to verify claims and flag unsupported information.

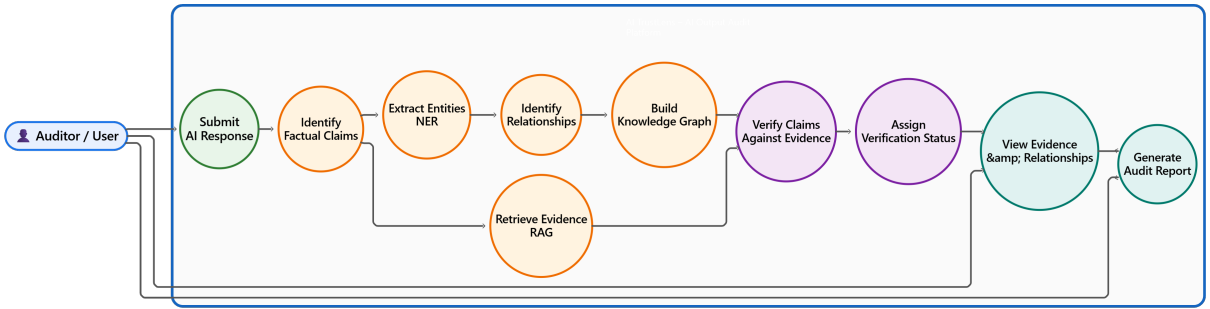
2. Objectives

- **Audit AI outputs** for factual accuracy and consistency.
- **Extract claims and entities** using claim identification and NER.
- **Integrate RAG and graph context** for evidence retrieval and relationship-aware verification.
- **Classify claims:** Supported, Refuted, Conflicting Evidence, or Not Enough Evidence.
- **Flag unsupported or inconsistent claims** with evidence-backed explanations.
- **Generate traceable audit reports:** claims, evidence, entities, relationships, and status.

3. Project Scope [1/2]
Key Points

1. **Input**
   - AI-generated textual responses submitted for auditing.
2. **Processing**
   - Claim identification, NER, and relationship extraction.
   - RAG evidence retrieval and graph-based claim verification.
3. **Output**
   - Claim status: Supported, Refuted, Conflicting Evidence, or Not Enough Evidence.
   - Traceable audit report with evidence, entities, and relationships.
4. **System Boundary**
   - Text-only auditing; assists verification without guaranteeing correctness.

3. Project Scope [2/2]
Auditing Workflow



Evidence-backed Fact Checking using RAG and Few-Shot In-Context Learning with LLMs

Ronit Singal et al.

- Retrieves the top three relevant documents for a given claim.
- Uses a large language model (LLM) to extract evidence from the retrieved documents.
- Predicts claim veracity through few-shot in-context learning, without fine-tuning.
- Classifies claims as **Supported**, **Refuted**, **Conflicting Evidence**, or **Not Enough Evidence**.

Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

Patrick Lewis et al.

- Combines a parametric generator with a non-parametric document retriever.
- Grounds generated output in retrieved evidence rather than relying on model memory alone.
- Achieved state-of-the-art results on open-domain question-answering tasks at the time of publication.
- Produces more specific, factual, and diverse text than generation-only baselines.

Language Models are Few-Shot Learners

Tom B. Brown et al.

- Introduces GPT-3, a language model with 175 billion parameters.
- Performs new tasks using a prompt and a few examples, without gradient updates.
- Established the in-context, few-shot prompting paradigm.
- Shows that large language models can generate fluent text that sometimes contains fabricated information.

LoRA: Low-Rank Adaptation of Large Language Models

Edward J. Hu et al.

- Freezes pre-trained model weights during adaptation.
- Injects small, trainable low-rank matrices into Transformer layers.
- Reports reductions of up to 10,000$\times$ in trainable parameters and 3$\times$ in GPU memory requirements compared with full fine-tuning.
- Matches or exceeds full fine-tuning quality on multiple benchmarks.

Survey of Hallucination in Natural Language Generation

Ziwei Ji et al.

- Surveys hallucination across summarization, dialogue, question answering, and translation.
- Defines **intrinsic hallucination** as generated content that contradicts the source.
- Defines **extrinsic hallucination** as generated content that cannot be verified against the source.
- Reviews existing evaluation metrics and methods for mitigating hallucination.

FEVER: a Large-scale Dataset for Fact Extraction and VERification

James Thorne et al.

- Introduces the FEVER dataset, containing 185,445 claims derived from Wikipedia.
- Claims are manually verified, with evidence sentences annotated for supported and refuted claims.
- Classifies claims as **Supported**, **Refuted**, or **Not Enough Information**.
- Established the retrieve → select evidence → classify pipeline for fact verification.

BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding

Jacob Devlin et al.

- Pre-trains a bidirectional Transformer using masked language modeling.
- Conditions each token's representation on both left and right context.
- Fine-tunes the model with just one additional output layer per task.
- Achieved state-of-the-art results on eleven NLP tasks at the time of publication.

Knowledge Graphs

Aidan Hogan et al.

- Presents a comprehensive survey of knowledge graph data models.
- Covers graph representation, querying, and schema.
- Discusses deductive and inductive knowledge over graphs.
- Provides a foundation for using graph databases to structure entities and their relationships.

5. Proposed System Design

- AI-generated medical response is decomposed into **atomic factual claims**.
- A **Medical NER model** extracts key entities (diseases, drugs, symptoms, treatments) and their relationships.
- Entities and relationships are stored as nodes and edges in a **Knowledge Graph (Neo4j)**.
- Supporting evidence is retrieved via a **hybrid approach**: RAG over trusted sources (PubMed, WHO) combined with graph-based context.
- Each claim is **verified** against the combined evidence and classified as Supported, Refuted, Conflicting Evidence, or Not Enough Evidence, with an evidence-backed explanation.

6. Methodology [1/4]
Overview

Six-stage pipeline

1. Data Collection
2. Claim Extraction
3. NER + Relation Extraction
4. Knowledge Graph Construction
5. Evidence Retrieval
6. Claim Verification

- Each stage feeds the next; the knowledge graph persists across runs.
- **Goal:** Reduce unsupported or hallucinated claims in AI-generated medical responses.

6. Methodology [2/4]
Data Collection & Claim Extraction

- **Trusted sources:** Collect PubMed abstracts, WHO materials, and a CORD-19-style corpus.
- **Audit input:** Collect AI-generated medical responses.
- **Claim extraction:** Use an LLM to decompose each response into atomic factual claims, following the approach of FActScore.
- **Output:** A list of checkable claims.

6. Methodology [3/4]
NER, Relation Extraction & Knowledge Graph Construction

- **Medical NER:** Use a BioBERT-based model to extract entities such as diseases, drugs, symptoms, and treatments.
- **Relation extraction:** Link entities through relationships, for example, Drug X $\xrightarrow{\text{treats}}$ Disease Y.
- **Graph construction:** Represent entities as nodes and relations as edges, stored in Neo4j.
- **Persistence:** Maintain a persistent graph that accumulates across documents.

6. Methodology [4/4]
Evidence Retrieval & Claim Verification

- **Hybrid retrieval:** Combine RAG text embedding search with graph queries for structural context.
- **Evidence combination:** Merge retrieved text and graph context into a combined evidence set.
- **Claim verification:** Classify each claim as **Supported**, **Refuted**, **Conflicting Evidence**, or **Not Enough Evidence**.
- **Explanation generation:** Generate an evidence-backed explanation that feeds into the audit report.

7. Evaluation Methodology [1/3]

Procedure

1. **Dataset:** Prepare CORD-19-derived medical claims and claims extracted from AI-generated medical responses.
2. **Ground truth:** Manually annotate claims against trusted medical literature as Supported, Refuted, Conflicting Evidence, or Not Enough Evidence; review annotation disagreements.
3. **Sample size and split:** Finalize the total claim count and development/test split before evaluation (**TBD**). Report per-class counts and keep source documents disjoint between splits.
4. Evaluate the **RAG-based approach** as the baseline.
5. Evaluate the proposed **NER + RAG + Graph** approach on the same held-out claims and evidence corpus, without access to test labels during prediction.
6. Compare both approaches at the claim level.

7. Evaluation Methodology [2/3]

Metrics

Metric	**Purpose**
Accuracy	Overall verification correctness
Macro-Precision	Mean of per-class precision across all four classes
Macro-Recall	Mean of per-class recall across all four classes
Macro-F1	Mean of per-class F1 scores; gives each class equal weight despite class imbalance
Hallucination Rate (miss rate)	Share of gold Refuted / Not Enough Evidence claims incorrectly predicted as Supported
Confusion Matrix	Four-class error breakdown, with per-class Precision / Recall / F1

Compute every metric for both systems on the same test set. Not Enough Evidence denotes insufficient evidence, not proven falsity.

7. Evaluation Methodology [3/3]

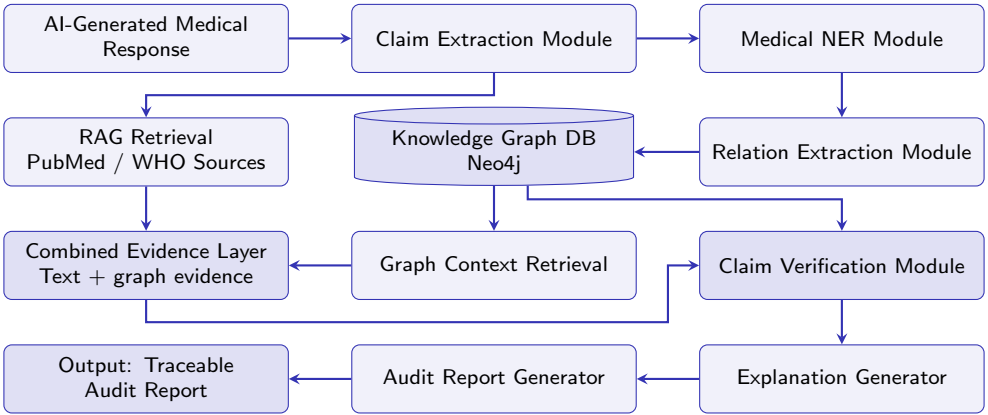
Baseline vs Proposed

Planned comparison on the held-out test set

Metric	**Baseline: RAG**	**Proposed: NER + RAG + Graph**
Accuracy ↑	TBD	TBD
Macro-Precision ↑	TBD	TBD
Macro-Recall ↑	TBD	TBD
Macro-F1 ↑	TBD	TBD
Hallucination Rate (miss rate) ↓	TBD	TBD

↑: Higher is better. ↓: Lower is better. **TBD**: To be measured; no experimental results are reported yet.

8. System Architecture



Claim status: Supported / Refuted / Conflicting Evidence / Not Enough Evidence

9. Feasibility Analysis [1/3]
Technical

The required technologies are available and can be integrated to develop the proposed system.

- **NLP / Claim Extraction**
  Existing NLP/LLM techniques can extract factual claims from AI responses.
- **Medical NER**
  Pre-trained models are available to identify medical entities (diseases, drugs, symptoms, treatments, etc.).
- **Evidence Retrieval**
  RAG and retrieval techniques can fetch relevant information from trusted sources.
- **Semantic Similarity**
  Pre-trained embedding models can compare claims with retrieved evidence.
- **Graph Database**
  Neo4j or similar databases can represent relationships between medical entities.
- **Explainable Output**
  Claim-wise results with supporting evidence can be generated using existing tools.

9. Feasibility Analysis [2/3]
Economic & Resources

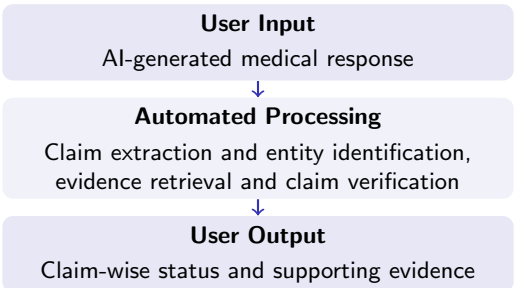
The project can be developed using available resources and open-source tools with manageable cost.

Resource	Availability / Approach
Programming	Python
NLP / NER Models	Pre-trained / Open-source models
Embedding Models	Existing pre-trained models
Graph Database	Neo4j / equivalent
Evidence Sources	Trusted medical sources (e.g., PubMed, WHO)
Datasets	Publicly available datasets
Hardware	Existing computer / lab resources (GPU where required)
Development Cost	Low-cost academic prototype (mostly open-source tools)

9. Feasibility Analysis [3/3]

Operational

The proposed workflow is designed to support user-submitted medical text auditing with evidence-backed results.



- Usability and response time will be evaluated during prototype testing.
- Results support human review; the system is not intended for diagnosis or treatment.

10. Overview of Existing System

- Existing systems perform automated claim/fact verification using NLP, LLMs and external evidence.
- The general verification pipeline includes claim processing, evidence retrieval, evidence selection, veracity prediction and explanation generation.
- Recent approaches use RAG to retrieve relevant documents from external knowledge sources before verification.
- The RAG and few-shot ICL system retrieves the top 3 relevant documents, extracts evidence using LLMs, and predicts the claim's veracity.
- It provides four outcomes: Supported, Refuted, Conflicting Evidence/Cherrypicking, and Not Enough Evidence.
- Key limitations: hallucinated evidence, irrelevant retrieved context, knowledge conflicts, and difficulty with complex or conflicting claims.

11. Existing System vs Proposed Approach

Existing Approach	**Proposed Approach**
Retrieves text/document evidence	Identifies key entities using NER.
Limited explicit relationship representation	Graph-based relationship representation
Context may be unstructured	Structured graph context
Higher possibility of unsupported responses	Aim to improve contextual grounding
Limited entity-level representation	NER-based entity identification

12. Conclusion

- The project aims to addresses unreliable or unsupported information in AI-generated responses.
- NER is used to identify important entities from input information.
- A graph database provides structured representation of entities and their relationships.
- Graph-based context retrieval can provide relevant information to support response generation.
- The proposed system will be further implemented, tested and evaluated to assess its effectiveness in reducing hallucinations.

13. References

- Ronit Singhal, Mohit Khandelwal, Ankur Sharma, *Evidence-backed Fact Checking using RAG and Few-Shot In-Context Learning with LLMs" (FEVER 2024)*, Conference, 2024.
- Ramy Baly, Mitra Mohtarami, James Glass, Lluís Màrquez, Alessandro Moschitti, and Preslav Nakov. *Integrating stance detection and fact checking in a unified corpus*, Conference, 2018.
- ames Thorne and Andreas Vlachos. *Automated fact checking: Task formulations, methods and future directions.*, Conference, 2018.
- Eldar Kurtić, Alexandre Marques, Shubhra Pandit, Mark Kurtz, and Dan Alistarh, *"Give Me BF16 or Give Me Death"? Accuracy-Performance Trade-Offs in LLM Quantization*, Conference, 2025.
- Anmol Mekala, Anirudh Atmakuru, Yixiao Song, Marzena Karpinska, and Mohit Iyyer (with Anmol Mekala and Anirudh Atmakuru contributing equally as co-first authors), *Does quantization affect models' performance on long-context tasks?*, Conference, 2025.

Thank You